Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			award (1) for number of moles of carbon **and** hydrogen $\frac{85.7}{12.0} = 7$ and $\frac{14.3}{1.01} = 14$ ratio 7:14 ⇨ 1:2 empirical formula CH_2 (1)		2		2		
	(b)			$n(NaOH) = \frac{26.40 \times 0.100}{1000} = 2.64 \times 10^{-3}$ (1) 2.64×10^{-3} mol of acid (in 25.0 cm^3) 1.06×10^{-2} mol of mol of acid (in 100 cm^3) (1) $M_r = \frac{0.93}{1.06 \times 10^{-2}} = 88$ (1)		3		3	2	3
	(c)			**A** but-1-ene (1) **B** butan-1-ol (1) **C** butan-2-ol (1) **D** butanoic acid (1) ecf possible e.g. from incorrect alkene			4	4		
	(d)			acidified (potassium) dichromate(VI)		1		1		1
				Question 9 total	**0**	**6**	**4**	**10**	**2**	**4**